MIL-R-22762 (AS)
Amendment 5
12 Aug 1968

Superseding:
Amendment 4
28 September 1967

MILITARY SPECIFICATION

RECEIVER-TRANSMITTER RADIO
RT-546/ASQ-19, RT-546A/ASQ-19, RT-853/ASQ-19 and RT-546B/ASQ-19

This amendment forms a part of Military Specification MIL-R-22762 (WEP) dated 20 February 1961 and has been approved by the Naval Air Systems Command, Department of the Navy.

By this amendment basic MIL-R-22762 (WEP) is changed to MIL-R-22762 (AS) and throughout the specification, Bureau of Naval Weapons shall be changed to Naval Air Systems Command.

Page 1, Title: Delete and substitute the title of this amendment.

Page 1, After paragraph 1.1(4): Add Notes 1 and 2.

"Note 1 The requirements for the RT-546A/ASQ-19 and RT-546B/ASQ-19 shall be the same as for the RT-546/ASQ-19 and RT-853/ASQ-19 respectively except that the operation of both main and guard receiver shall be from a common antenna input.

"Note 2 The RT-853/ASQ-19 and RT-546B/ASQ-19 are compatible with Juliet-28 equipment."

Page 1, Paragraph 1.2: Add

"Juliet-28 (RT-853 and RT-546B only)
Control C-8057/ARC (RT-853 and RT-546B only)"

Page 1, Paragraph 2.1, after "invitation for bids:" Add

"except the dated document indicated below,"

Page 2, Paragraph 2.1, under SPECIFICATIONS Military: Add

"MIL-I-6181B (29 May 1953)	Interference Limits, Tests and Design Requirements, Aircraft Electrical and Electronic Equipment (See 2.2.(2))"

Page 2, Paragraph 2.2: Add

"(2) Specification MIL-I-6181B is available from Naval Air Systems Command, Washington 25, D. C. Attention **Avionic Division.**"

FSC 5821

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Page 2, Paragraph 3.2.2: Delete "45.6"and substitute: "47.0"

Page 2, Add

"3.2.8 Spurious Radiation - The magnitude of all radiated signals on other than the carrier frequency shall be 40 db or more below the carrier signal except that harmonic frequencies shall be attenuated not less than 55 db at 450 mc with a linear increase to 75 db at 525 mc. The attenuation above 525 mc shall be 75 db or greater. This requirement shall apply when the low pass filter described in 3.4.2 of this specification is connected in series with the transmission line.

3.2.8.1 Antenna Conducted Interference - Antenna conducted interference shall meet the requirements of Specification MIL-I-6181B except that at not more than 9 of the 18 injection frequencies the antenna conducted interference due to the main receiver injection frequency shall not exceed 300 microvolts."

*Page 3, Paragraph 3.3.2: Add

"The sensitivity of the auxiliary audio and the auxiliary guard audio shall be such that when an R-F input of 5 microvolts (open circuit) modulated 30% at 1000 cycles is applied, the signal plus noise-to-noise ratio shall be not less than 3.0 db for the auxiliary audio and not less than 6 db for the auxiliary guard audio."

Page 3, Paragraph 3.3.2.1: Delete and substitute

"3.3.2.1 Degraded or Unusable Channels Due to Internal Interference -

Certain main receiver channel frequencies may be degraded or unusable due to internally generated interference. When the equipment is supplied with the guard frequency set for 243.0 mc, the main receiver sensitivity measurements shall be made with an input of 20 microvolts at frequencies 241.1, 241.2, and 275.6 mc. Six unusable channels 242.1, 246.6, 253.3, 287.1, 322.8 and 339.0 mc are permitted which are independent of the guard receiver frequency."

Page 3, Paragraph 3.3.4: Delete table and substitute

"ATTENUATION BELOW RESONANT RESPONSE	TOTAL BANDWIDTH		
	GUARD RECEIVER	RT-546/RT-546A MAIN RECEIVER	RT-853/RT-546B MAIN RECEIVER
6 db	60 Kc min	60 Kc min	80 Kc min
60 db	400 Kc max	200 Kc max	200 Kc max'

Page 4, Paragraph 3.3.9.1: Add

"The auxiliary guard output shall be 0.15-0.45 volt rms when loaded with 600 ohms."

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Page 4, Paragraph 3.3.9.3: Delete and substitute

"3.3.9.3 Auxiliary Circuit - The auxiliary circuit shall deliver only signals detected by the variable frequency receiver. When a 1000 - microvolt signal modulated 30% at 1000 cps is applied, the output shall be at least 0.25 volt, and when the modulation frequency is varied from 70 to 7000 cps, the output for any frequency in the range 70 to 200 cps shall be within +4.5 and -3.0 db and in the range 200 to 7000 cps shall be within +2 and -3 db of the output obtained at 1000 cps. The RT-853 and RT-546B shall not vary more than +4 and -4 db from the output obtained at 1000 cps when modulation frequency is varied from 70 to 20,000 cps."

Page 6, Paragraph 3.3.12.1.1: Delete table and substitute

"Frequency cps	Relative Output in db RT-546/RT-546A	RT-853/RT-546B
150	-8 or lower	-5 or lower
300	-3 to -7	+4 to -4
600	+1 to -2	+3 to -3
1,000	0 (ref. level)	0 (ref. level)
3,000	+2 to -2	+2 to -2
6,000	+2 to -2	+3 to -3
10,000	+1 to -5	+4 to -4
15,000	--	+4 to -4
20,000	--	+4 to -4

Page 6, Paragraph 3.3.12.2: Delete and substitute

"3.3.12.2 Tone Modulation - Not less than 70 percent modulation of the r-f carrier shall occur when tone transmission is selected and the transmitter is keyed. The frequency of modulation shall be 1000 cps ± 20%."

*Page 6, Paragraph 3.3.15: Delete and substitute

"3.3.15 Noise Modulation - Under standard conditions, noise modulation of the transmitter carrier shall be at least 40 db below the maximum modulation obtainable and the receiver signal plus noise-to-noise ratio shall be at least 30 db when a 10,000-microvolt signal modulated 30% at 1000 Hz is applied and the total receiver output (Signal plus noise) is adjusted to be 250 milliwatts. Under the above condition the auxiliary audio and the auxiliary guard audio shall have a signal plus noise-to-noise ratio of not less than 20 db."

Page 8, Paragraph 3.4.1.2: Delete "7.52" and substitute "7.75".

Page 10, Paragraph 3.4.1.7.3: Add

"The auxiliary guard output of the RT-853 and RT-546B shall be terminated in a 600ohm load, including means for measuring the audio output voltage."

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Page 10, Paragraph 3.4.1.7.4, under "Carbon Microphone Input" and under "Dynamic Mike Input": Delete

"Frequency range 150-10,000 cps" and substitute
"Frequency range 150-10,000 cps (RT-546 and RT-546A)
150-20,000 cps (RT-853 and RT-546B)"

Add Note: 1 Volt RMS open circuit must be supplied when the RT-853/ASQ-19 or RT-546B/ASQ-19 is used with Juliet equipment".

Page 11, Paragraph 3.4.1.8: Delete and substitute

"3.4.1.8 Presurization - The case containing this equipment requires a supply of dry pressurized air. At an internal pressure of 15 psi gauge, the case construction shall be such that the leakage is no greater than 0.5 psi in 2 hours. The equipment performance shall be in accordance with this specification when the internal pressure is maintained at 20 $\pm$ 5 psia."

*An asterisk preceding a page number denotes a change to be made to the specification which did not appear in Amendment 4.

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